Multi-Task Learning for Stock Price Prediction

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Abstract

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I. Introduction

Insert introduction here

II. Motivation and Objectives

Motivation

Objectives

III. Methodology

A. Data

- Companies
- aggregation choice
- time choice

B. Algorithms

Random Forests:

LSTMS:

C. Preprocessing

Misaligned Dates: Due to different market holidays, some days in the data included both null values and recorded prices. The days with null values were removed to keep the dataset consistent. Dropping a few days of data should not obscure the stocks' growth patterns and relationships.

Logarithmic Scale: The data was scaled logarithmically. This was to ensure stable predictions by reducing effect of outliers on the model and to account for the exponential patterns in each stock's data. The predictions become more stable because most analysis techniques assume additive data, including machine learning. [1].

The scaling affect is shown in appendix A, the medium stock ACIW is on a similar scale to the large stock AAPL after logging the data.

Random Forests: describe script and set up of what it intakes [1]

LSTMS: For

D. Building the Models (?)

Random Forests:

LSTMS:

IV. Results

add pics and stuff

A. Hyperparameters

B. Algorithm Performances

V. Discussion

A. Findings

B. Interpretation of Results

C. Hyperparameters

D. Advantages and Disadvantages of Multi-Tasking

VI. Conclusion

References

[1] "Decomposition of time series," 2025. [Online]. Available: https://medium.com/@m.capurso/decomposition-of-time-series-071ce7b7a5a0

Appendix

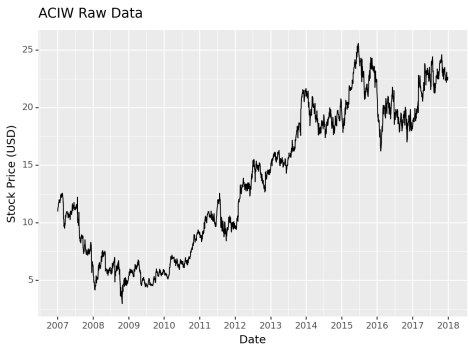

Fig. A1. Raw price data of ACIW Stock

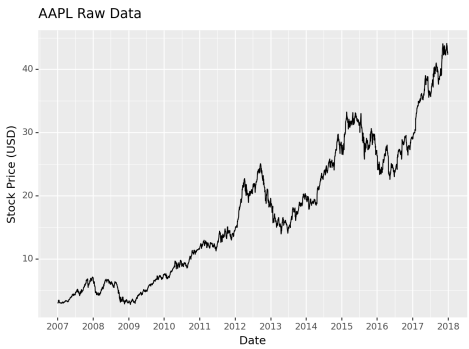

Fig. A2. Raw price data of APPL Stock

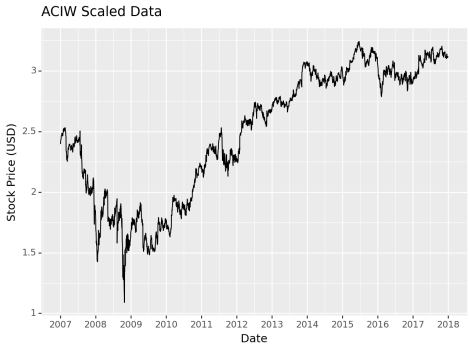

Fig. A3. Logged price data of ACIW Stock

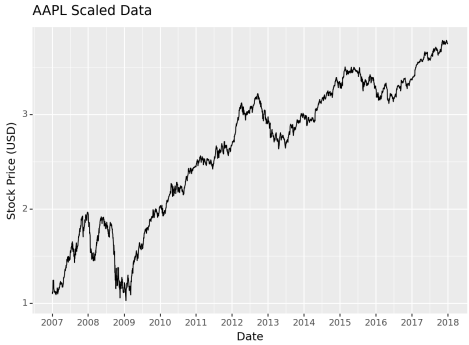

Fig. A4. Logged price data of APPL Stock